

Advanced (Habanero)

Q1. Solve for x : $2^x = 8$

- (A) $x = 2$
- (B) $x = 3$
- (C) $x = 4$
- (D) $x = 5$
- (E) $x = 6$

Q2. Solve for y : $3^y = 27$

- (A) $y = 1$
- (B) $y = 2$
- (C) $y = 3$
- (D) $y = 4$
- (E) $y = 5$

Q3. Solve for z : $4^z = 16$

- (A) $z = 1$
- (B) $z = 2$
- (C) $z = 3$
- (D) $z = 4$
- (E) $z = 5$

Q4. Solve for x : $x^2 = 4^3$

- (A) $x = 4$
- (B) $x = 6$
- (C) $x = 8$
- (D) $x = 10$
- (E) $x = 12$

Q5. Solve for y : $2^{y+1} = 32$

- (A) $y = 3$
- (B) $y = 4$
- (C) $y = 5$
- (D) $y = 6$
- (E) $y = 7$

Q6. Solve for z : $3^{z-1} = 9$

- (A) $z = 1$
- (B) $z = 2$
- (C) $z = 3$
- (D) $z = 4$
- (E) $z = 5$

Q7. Solve for x : $x^3 = 8^2$

- (A) $x = 2$
- (B) $x = 4$
- (C) $x = 6$
- (D) $x = 8$
- (E) $x = 10$

Q8. Solve for y : $y^2 = 3^4$

- (A) $y = 3$
- (B) $y = 6$
- (C) $y = 9$
- (D) $y = 12$
- (E) $y = 15$

Open Ended Questions (FRQ)

Q9. Solve for x : $2^{x+2} = 128$

Q10. Solve for z : $z^2 = 2^{2z}$

Chef's Tips

- Read each question carefully
- Use the back of the page for calculations
- Chef's Tips